REPORT NO. 02: CONTRIBUTION TO SRS (CHAPTER 1 AND 2)

COURSE CODE: CSE 314
COURSE TITLE: OBJECT ORIENTED ANALYSIS AND DESIGN LABORATORY

Submitted by

Tanvir Alam Shihab (334)

Murad Hossain Rohan (338)

K.M. Taskif Bin Parves (343)

Naosin Tabassum Lineya (364)

Marufa (365)

Submitted to

Dr. Md. Musfique Anwar, Professor

Sarnali Basak, Associate Professor

Computer Science and Engineering

Jahangirnagar University

Dhaka, Bangladesh

February 09, 2026

Contents

Abstract **iv**

1 OBJECTIVE **1**

2 EQUIPMENT USED **2**

3 SRS DOCUMENT **3**

3.1 Chapter 1 3

3.1.1 Purpose 3

3.1.2 Intended Audience 4

3.1.3 Intended Use 4

3.1.4 Product Scope 5

3.1.5 Risk Definition 6

3.2 Chapter 2 8

3.2.1 User Classes and Characteristics 8

3.2.2 User Needs 9

3.2.3 Operating Environment 10

3.2.4 Constraints 11

3.2.5 Project Assumption 12

4 INDIVIDUAL CONTRIBUTIONS **13**

4.1 My Contribution 13

4.2 Evidence of Contribution 13

4.2.1 Trello Task Board 13

4.2.2 Discord Communication 14

4.3 SRS Section 15

4.3.1 1.5 Risk Definition 15

4.3.2 2.5 Assumptions 16

5 DISCUSSION AND PROOFS **16**

5.1 Trello Activity Proof 16

5.2 Discord Communication Proof 17

6 CONCLUSIONS **18**

Abstract

This report presents the analysis and documentation of a Multi-Tenant E-Commerce System based on the Software Requirements Specification (SRS). The purpose of this work is to define the intended use, product scope, target users, and operating environment of the proposed system. The system is designed to support small and social-media-based businesses by providing a unified digital platform for online storefront management and wholesale product sourcing. Through requirement analysis, the report outlines how the system addresses operational inefficiencies, improves accessibility to digital commerce, and ensures scalability through a cloud-based, multi-tenant architecture.

1. OBJECTIVE

The main objectives of this lab are:

1. To understand the role of an SRS in defining software requirements clearly and systematically.
2. To identify the intended users and use cases of a multi-tenant e-commerce platform.
3. To analyze the product scope, operating environment, and system constraints at an early design stage.
4. To document functional expectations and environmental requirements without focusing on implementation details.
5. To ensure alignment between business goals and technical requirements through structured requirement analysis.

2. EQUIPMENT USED

The following tools and resources were used to complete this lab:

1. Personal computer or laptop.
2. Web browsers (Chrome, Firefox, Brave)
3. Word processing softwares for individual parts of the SRS (Notepad, Text Editor, MS Word)
4. Software Requirements Specification (SRS) guidelines and templates (Given in the Google classroom)
5. Discord server for group interraction and discussion
6. Trello for managing project tasks
7. Course materials related to Object-Oriented Analysis and Design (OOAD)

3. SRS DOCUMENT

3.1 Chapter 1

Introduction

The rapid growth of digital commerce has transformed the way small businesses operate, with many entrepreneurs relying heavily on social media platforms such as Facebook and WhatsApp to reach customers. While these platforms are effective for marketing and communication, they lack essential features for structured order management, inventory control, and business analytics. As a result, small vendors often struggle to scale their operations efficiently.

The Multi-Tenant E-Commerce System is proposed as a Software as a Service (SaaS) platform designed to bridge this gap. The system enables multiple vendors to register and instantly obtain their own personalized online storefront without the need for technical expertise. By employing a multi-tenant architecture, the platform ensures efficient resource usage while keeping each vendor's data isolated and secure. This solution empowers small business owners to manage their products, customers, and orders professionally, while significantly reducing the cost and complexity of building and maintaining individual e-commerce websites.

3.1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements for the Multi-Tenant E-Commerce System. This document outlines the architectural constraints and behavioral specifications necessary to build a "Software as a Service" (SaaS) platform that empowers small vendors, specifically Social Commerce entrepreneurs (operating on platforms like Facebook and Instagram), to launch dedicated online storefronts without technical expertise.

The primary objective of the system described in this document is to automate the man-

ual order-taking process currently used by small vendors. By leveraging a multi-tenant architecture, the system allows a ”Super Admin” to host a single application instance that serves multiple ”Tenants” (Vendors), providing each with a unique URL, rigorous data isolation, and a personalized storefront experience.

This document serves as the authoritative source for the system’s requirements and validation criteria. It is intended for:

- **Developers:** To understand the functional specifications, data isolation requirements, and user interface constraints necessary for development.
- **Project Supervisors & Reviewers:** To verify that the proposed system meets academic standards and fulfills the objectives outlined in the project proposal.
- **Quality Assurance:** To define the testing criteria and expected system behaviors for user acceptance testing.

3.1.2 Intended Audience

The primary intended audience of the Multi-Tenant E-Commerce System is small business owners, new entrepreneurs, and F-commerce sellers who primarily operate through platforms like Facebook and WhatsApp. These individuals often lack the financial resources and technical knowledge required to develop and maintain a full-scale e-commerce website. This system is designed to support vendors who want to professionalize their business, manage orders and products more efficiently, and build a stronger online presence without the burden of high development costs. Additionally, the platform can benefit freelancers, home-based businesses, and local sellers who wish to expand digitally with minimal investment.

3.1.3 Intended Use

The Multi-Tenant E-Commerce System is intended to serve as a cloud-based digital commerce platform that supports small and medium-sized businesses in managing and growing their online operations. The system is designed primarily for sellers who currently rely on social media platforms such as Facebook, Instagram, and WhatsApp for conducting business but lack structured tools for managing customers, orders, and inventory.

The platform will be used by business owners to create and manage their own branded online storefronts, where they can display products, track stock levels, process customer

orders, and monitor sales activity. In addition, the system will provide integrated access to verified wholesale suppliers, enabling sellers to restock products directly through the same platform without needing to search for suppliers separately.

Wholesale suppliers will use the system to list products in bulk, manage available inventory, and receive purchase orders from registered sellers. End customers will interact with the platform indirectly through individual seller storefronts, where they can browse products, place orders, and complete purchases.

The system is intended to be accessed through standard web browsers and mobile devices, ensuring usability for individuals with limited technical knowledge and primarily mobile-based internet access. Overall, the intended use of the system is to simplify digital commerce operations, reduce manual workload, and create a more organized and scalable business environment for small online sellers.

3.1.4 Product Scope

So, the Multi-Tenant E-Commerce System is basically a cloud-based system which is supposed to allow small business owners to get a leg up, especially those who are kinda floundering with social media trying to find and deal with customers and sales. We're trying to give them one digital storefront.It's an instructable pan in which a business owner can post stuff, keep track of inventory, ship orders, all under their own branded online shop.

Benefits:

For Small Businesspersons (Tenants):

- **Customer Acquisition & Retention:** Among others, it transcends the limitations of Instagram, Facebook or TikTokś algorithms as it could offer a polished and dedicated online store that increases discover-ability and credibility.
- **Operational Efficiency:** Sweeps up all questions, orders, payments by customers - Facebook, Insta, WhatsApp - into one tidy dashboard.
- **Reduced administrative Hassle:** Chess (like tracking inventory, tracking orders, and rudimentary sales reports) occurs automatically.
- **Low Barrier to Entry:** Provides an affordable, scalable solution to creating a custom e-com platform, and you don't need to be a tech wizard.

Objectives & Goals:

1. **Primary Objective:** Pump up sales and reduce operational headaches for small, social media heavy businesses by launching an all in one e-commerce platform.
2. **Technical Goal:** Developing a safe, scalable and reliable multi-tenant architecture in which the data of each tenant is completely isolated - but the entire system is maintained and updated from a single codebase.
3. **User Experience Goal:** Deliver a snappy, mobile-first interface that doesn't need hours of training. Let's say that your tenant wants to get their store up and running in a few hours.

Overall Business Goals: This SRS document and resultant software play an important role in hitting the organization's strategic business targets:

- **Market Penetration:** Provides an obvious product to acquire a sizeable chunk of the non-serve segment of digitally-natives in small-business owners.
- **Brand Establishment:** Positions us as a go-to facilitate small-biz digital transformation in the region.
- **Cross-Functional Alignment:** This SRS provides context to teams that are outside of development:

Marketing & Sales: Use the features and benefits to build targeted marketing campaigns, sales collateral and explain the value for both tenants and suppliers.

Customer Support: Refer to the system's capabilities and user goals in building training materials, help docs, and supporting protocols.

Leadership/management: How to use the stated objectives and goals to track the success of the project and to steer future investments and decisions about the road map.

Finally, the scope of this product is to create a simple yet complex platform which reverses the way small social-centric businesses operate, which directly pushes the company's mission of expanding access to digital commerce as well as creating a sustainable SaaS business model.

3.1.5 Risk Definition

A risk is any potential issue that may negatively impact the system's security, legality, performance, or user trust during development or operation.

1. **Illegal Product Listing Risk**
   Vendors may attempt to sell illegal products such as drugs, weapons, or restricted items through the platform.
   Impact: Legal issues, platform misuse, and damage to system credibility.

2. **Fake or Invalid Payment Risk**
   Customers may attempt to make payments using fake, invalid, or fraudulent money or payment methods.
   Impact: Financial loss and disputes between vendors and customers.

3. **Payment Dispute Risk**
   Conflicts may arise regarding payment confirmation, delayed payments, or incomplete transactions.
   Impact: Customer dissatisfaction and trust issues with the platform.

4. **Product Return and Refund Dispute Risk**
   Disagreements may occur between vendors and customers regarding product returns, refunds, or damaged goods.
   Impact: Increased complaints and operational complexity.

5. **Data Security and Privacy Risk**
   There is a risk of unauthorized access to vendor or customer data if proper security controls are not implemented.
   Impact: Data breaches and loss of user trust.

6. **Vendor Misuse Risk**
   Some vendors may misuse the platform by providing false product information or misleading customers.
   Impact: Reduced reliability and customer confidence.

7. **User Adoption Risk**
   Vendors may prefer continuing business through social media instead of adopting a new platform.
   Impact: Low platform usage and limited success.

8. **Fraudulent Order and Cancellation Risk**
   Users may place false or spam orders, such as booking Cash on Delivery (COD) items with no intention of accepting delivery, or repeatedly creating and canceling orders to disrupt operations.
   Impact: Financial loss due to wasted shipping costs, inventory mismanagement (locking stock from real buyers), and increased operational overhead.

3.2 Chapter 2

3.2.1 User Classes and Characteristics

The Multi-Tenant E-Commerce System interacts with three distinct user classes. These classes are differentiated by their access levels, technical proficiency, and frequency of interaction with the platform.

Super Admin (System Operator)

The Super Admin is the primary stakeholder responsible for the maintenance and business viability of the SaaS platform. This user does not engage in commerce but rather facilitates the ecosystem for Tenants.

Characteristic	**Description**
Role	System Administrator and Business Owner.
Responsibilities	Monitoring server health, managing tenant subscriptions, enforcing platform policies (banning illegal items), and resolving high-level disputes.
Technical Proficiency	High. Expert understanding of web technologies, dashboard analytics, and administrative configurations.
Frequency of Use	Continuous. Daily access is required to monitor system stability and security logs.
Education/Background	Typically IT professionals or business administrators with experience in SaaS management.

Table 3.1: Super Admin Characteristics

Tenant (Vendor)

The Tenant is the core user of the system—a small business entrepreneur utilizing the platform to professionalize their existing S-commerce (Social Commerce) operations.

Characteristic	Description
Role	Storefront Manager and Seller.
Responsibilities	Creating the store, uploading products, managing inventory, processing orders, and handling customer inquiries.
Technical Proficiency	Low to Moderate. Highly proficient with mobile apps (Facebook/Instagram/WhatsApp) but lacks formal IT training or experience with complex web CMS platforms.
Frequency of Use	High/Daily. Constantly logged in via mobile devices to track sales and stock levels.
Education/Background	Varied background (home-based business owners, students, local traders). They require a low-code/no-code interface.

Table 3.2: Tenant (Vendor) Characteristics

End Customer (Shopper)

The End Customer is the external user who purchases goods. They view the Tenant's storefront as an independent brand and may be unaware of the underlying SaaS provider.

Characteristic	Description
Role	Buyer / Consumer.
Responsibilities	Browsing catalogs, adding items to cart, providing shipping details, and making payments.
Technical Proficiency	Varied. Ranges from digital natives to novice internet users. The interface must be intuitive enough for users with minimal digital literacy.
Frequency of Use	Variable. Usage is sporadic, driven by marketing campaigns or specific purchasing needs.
Education/Background	General public.

Table 3.3: End Customer Characteristics

3.2.2 User Needs

The Multi-Tenant E-Commerce System is designed to fulfill the needs of different types of users, including vendors, customers, and system administrators. The system aims to provide an easy-to-use, secure, and efficient platform for managing online business operations.

3.2.2.1 Vendor Needs

Vendors require a platform that allows them to easily create and manage their own online storefront without technical knowledge. Vendors need the ability to add, update, and remove products, manage inventory, track orders, and communicate with customers. They also require secure access to their business data and a simple interface for monitoring sales and order status. Finding and reaching customers easily without extra hassles of different social media platforms is also a necessary requirement.

3.2.2.2 Customer Needs

Customers need a convenient and reliable platform to browse products from different vendors, view product details, place orders, and track purchase status. They require secure payment options, clear product information, and a user-friendly interface to ensure a smooth shopping experience.

3.2.2.3 Administrator Needs

System administrators need tools to monitor platform activities, manage vendor registrations, and ensure compliance with system policies. They require the ability to monitor transactions, control system security, and prevent illegal or inappropriate product listings. Administrators also need to maintain system stability and manage user accounts when necessary.

3.2.3 Operating Environment

The Multi-Tenant E-Commerce System will operate as a cloud-based web application hosted on a scalable server infrastructure and delivered through a single system deployment. All vendors will use the same core frontend and backend system, ensuring consistency, maintainability, and efficient updates, while still allowing individual tenants to apply customizable themes, branding, and configuration settings to personalize their storefronts. The platform will be accessible through modern web browsers on desktops and mobile devices without requiring any additional software installation and will support common operating systems such as Windows, Linux, Android, and iOS. A stable internet connection will be required, as all business data, including orders, inventory, and product tracking information, will be processed and stored on the cloud servers. The system will include product tracking functionality that enables vendors

and customers to monitor order status and shipment progress through integrated logistics or tracking services. The cloud-based operating environment is designed to support secure communication, third-party integrations such as payment gateways and delivery providers, and scalable operation as the number of tenants and transactions grows.

3.2.4 Constraints

This section describes the predefined design, implementation and operational boundaries that the Multi-Tenant E-Commerce System needs to follow. These constraints are non-negotiable, and comprise the basic framework for the project.

3.2.4.1 Architectural Constraints

Multi-Tenant SaaS Model: The system must be coded and deployed as single and shared Software-as-a-Service (SaaS) application for multiple independent vendors (tenants). A monolithic or single tenant architecture is strictly prohibited.

Data Isolation: The system has to support strict logical data isolation between tenants at the database level. Data from one vendor should be totally inaccessible and invisible to all the other vendors under normal application usage. Physical database separation per tenant is not allowed because of scalability and maintenance overhead.

Shared Application Instance: All tenants shall be served from the same application codebase and server instance. Separate codebases or instances of servers for individual vendors are not permitted.

3.2.4.2 Design and Implementation Limitations

Super Admin Need: A system must have a special ”Super Admin” portal to manage the platform, including supervising tenant onboarding, platform-wide configuration, and system health monitoring.

Instant Tenant Onboarding: After successful registration and subscription, a new vendor must be given instant access into a full featured, personalized admin panel with a live, publicly accessible storefront URL. For standard onboarding flow manual configuration or setup delays for Super Admin are not allowed.

No Custom Code per Tenant: The system should not support tenant-specific custom code deployment. All the customization for a vendor’s storefront (e.g. colors, logo,

banner) is required to be done via configuration settings and theme options within the shared application.

Order and shipment boundary: As this platform is only for Bangladeshi tenant so order and delivery is not allowed in abroad. It has to be used within Bangladesh.

Preset Payment Gateways: Vendor storefronts are required to integrate with a predefined and limited set of payment gateways (e.g. bKash, Nagad) at the platform level. The system will not support the integration of arbitrary, vendor specific payment processors.

3.2.5 Project Assumption

Assumptions are conditions that are believed to be true for the successful development and operation of the system.

1. **Internet Availability Assumption:** It is assumed that vendors and customers will have access to a stable internet connection to use the platform.
2. **Legal Business Usage Assumption:** It is assumed that vendors will use the platform only to sell legal and permitted products and will follow local laws and regulations.
3. **Vendor Responsibility Assumption:** It is assumed that vendors are responsible for managing their own products, prices, inventory, and order fulfillment.
4. **Payment System Reliability Assumption:** It is assumed that integrated payment methods will function correctly and provide accurate transaction information.
5. **User Honesty Assumption:** It is assumed that vendors and customers will provide correct and truthful information during registration and transactions.
6. **Basic Technical Knowledge Assumption:** It is assumed that vendors have basic knowledge of using smartphones or web applications.
7. **System Usage Policy Assumption:** It is assumed that users will follow the platform's terms, conditions, and usage policies.
8. **Single Platform Maintenance Assumption:** It is assumed that the system will be maintained and updated by the platform administrator.
9. **Browser Compatibility Assumption:** It is assumed that users will access the system using modern web browsers.

4. INDIVIDUAL CONTRIBUTIONS

This chapter describes my individual contribution to the project work. Throughout the activity, I actively participated in planning, coordination, documentation, and report preparation. I used Trello to track tasks and Discord to communicate with team members and ensure smooth collaboration.

4.1 My Contribution

My responsibilities and contributions in this project include:

- Participated in team planning and discussion sessions.
- Completed my assigned documentation tasks.
- Helped organize and structure the final report.
- Managed task tracking using Trello.
- Communicated regularly with team members through Discord.
- Ensured all assigned work was completed before submission.
- Reviewed and edited report sections for clarity and formatting.

4.2 Evidence of Contribution

4.2.1 Trello Task Board

Trello was used to assign and track tasks during the project. The screenshots below show my assigned tasks and progress.

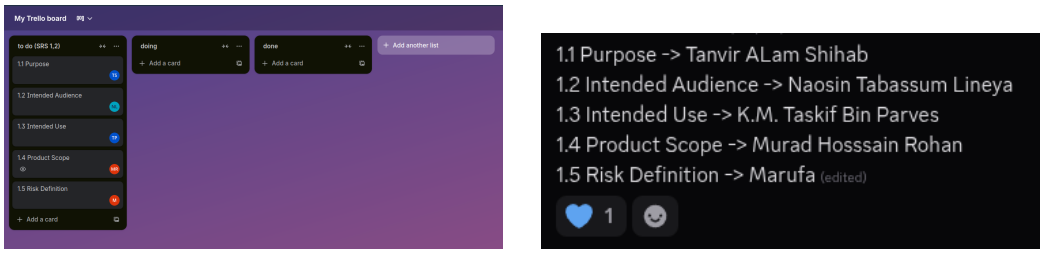

Figure 4.1: Trello board showing my task assignments and overall progress

4.2.1.1 Doing Section

The images below show the **Doing** column of the Trello board, representing tasks currently in progress.

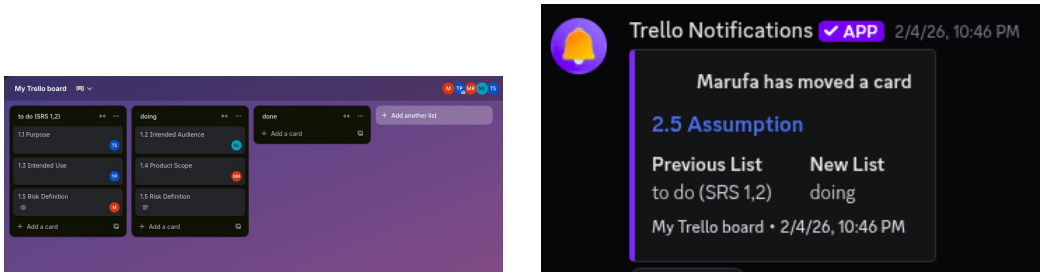

Figure 4.2: Trello “Doing” column showing tasks in progress

4.2.1.2 Done Section

The images below show the **Done** column of the Trello board, representing completed tasks.

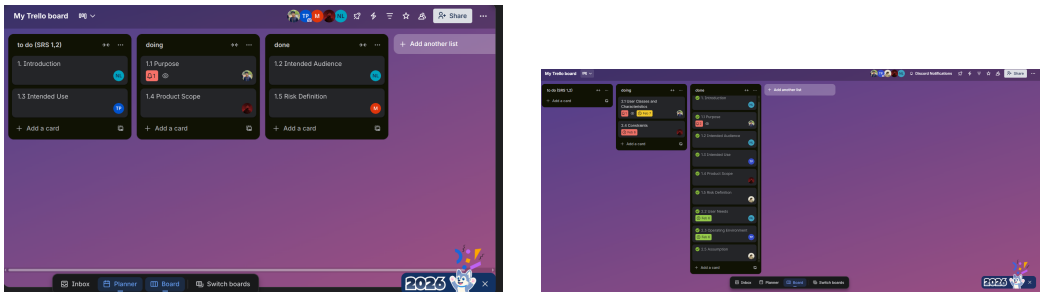

Figure 4.3: Trello “Done” column showing completed tasks

4.2.2 Discord Communication

Discord was used for team discussion and coordination. The images below show my participation in team communication and decision-making.

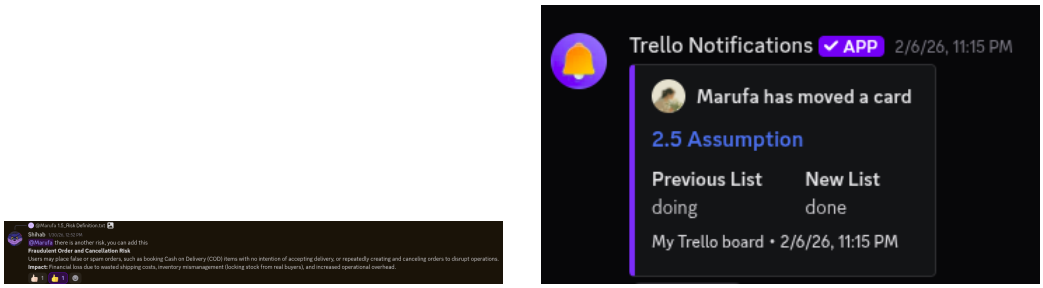

Figure 4.4: Discord communication showing my participation in discussions

4.3 SRS Section

4.3.1 1.5 Risk Definition

The following risks were identified during the project:

- **Communication Risk:** Miscommunication among team members could delay task completion.
- **Time Management Risk:** Delay in finishing assigned work could affect report submission.
- **Technical Risk:** Internet or device issues during documentation and coordination.
- **Coordination Risk:** Difficulty aligning schedules for discussion.

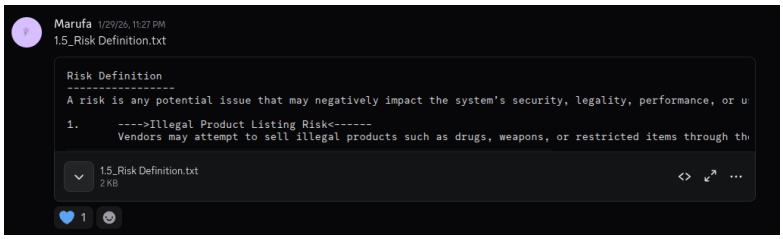

Figure 4.5: Risk Definition

Risk Mitigation:

- Regular communication through Discord.
- Task tracking and deadline monitoring using Trello.
- File sharing through cloud storage for backup.

4.3.2 2.5 Assumptions

The following assumptions were made during the project:

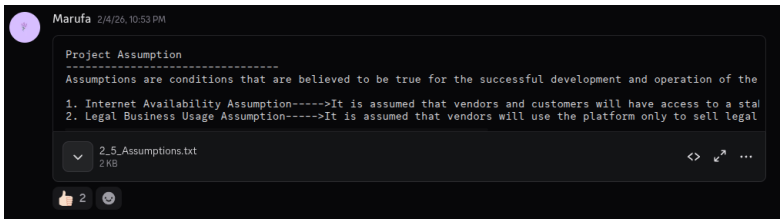

Figure 4.6: Assumptions

- All team members had access to internet and devices.
- Each member would complete assigned tasks responsibly.
- Communication platforms would remain accessible.
- Project files would be shared and stored properly.
- The report would be completed collaboratively and submitted on time.

5. DISCUSSION AND PROOFS

This chapter presents discussion and visual proof of the activities performed during the preparation of the Software Requirements Specification (SRS). Trello was used for task management and progress tracking, while Discord was used for communication, discussion, and file sharing.

These tools helped maintain organization, ensured timely completion of tasks, and allowed proper coordination during the preparation of Section 1.5 (Risk Definition) and Section 2.5 (Assumptions).

5.1 Trello Activity Proof

Trello was used to manage tasks related to the SRS report. Tasks were divided into **To Do**, **Doing**, and **Done** stages. The screenshots below show task progress, completion

status, and activity logs.

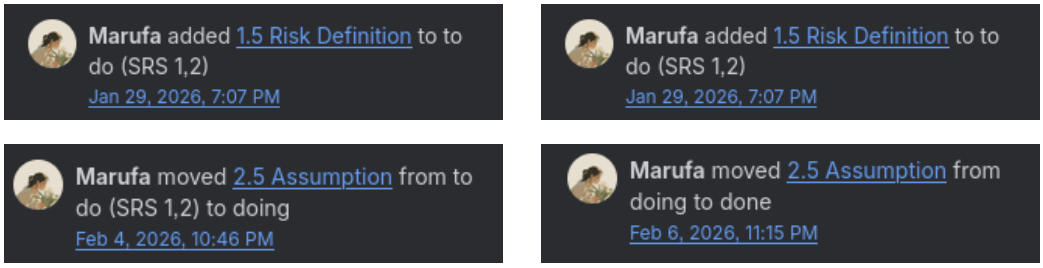

Figure 5.1: Trello board showing task planning, progress tracking, and completion

5.2 Discord Communication Proof

Discord was used for communication, discussion of SRS sections, and file sharing among team members. The following screenshots show task assignment messages, notifications, and document submission discussions.

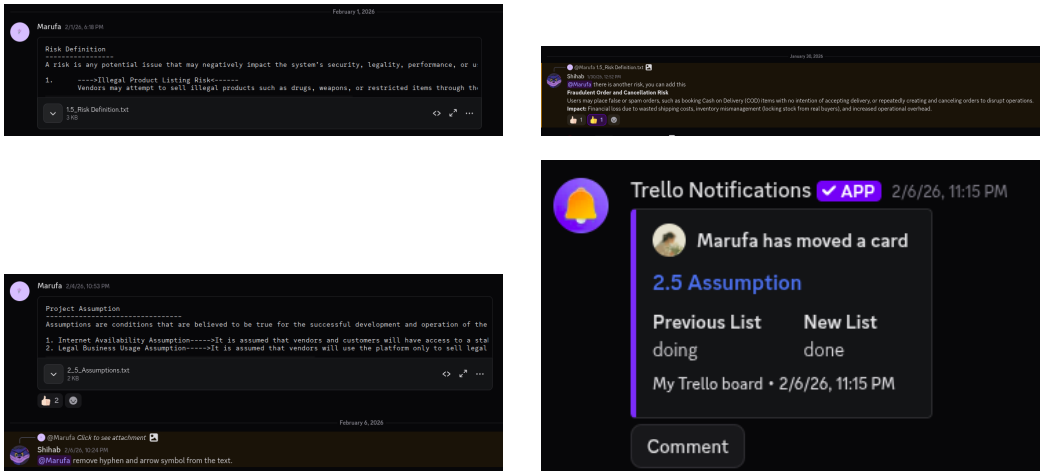

Figure 5.2: Discord discussion, task assignment, notifications, and file sharing

6. CONCLUSIONS

This lab activity successfully documented the foundational requirements of a Multi-Tenant E-Commerce System by analyzing the SRS (Chapter 1 and 2). The study clarified the system's purpose, intended users, and operating environment, emphasizing its role in supporting small and social-media-driven businesses. By defining the product scope and environmental constraints early, the SRS helps reduce ambiguity and provides a shared understanding among stakeholders. This requirement-focused approach ensures that future design and development phases can proceed in a structured and aligned manner, increasing the likelihood of delivering a scalable, secure, and user-friendly software solution.